



TEHNIČKO VELEUČILIŠTE
U ZAGREBU UNIVERSITY OF
APPLIED SCIENCES ZAGREB



**Stručni prijediplomski
studij Elektrotehnika**

3. semestar

**Komunikacijska i
računalna tehnika**

Ak.g. 2025./2026.

Signali i procesi

Fourierov red u kompleksnom obliku

Kalendar nastave po tjednima (ver.2)



#	Ponedjeljak	Utorak	Srijeda	Četvrtak	Petak
1	06.10.2025	07.10.2025	08.10.2025	09.10.2025	10.10.2025
2	13.10.2025	14.10.2025	15.10.2025	16.10.2025	17.10.2025
3	20.10.2025	21.10.2025	22.10.2025	23.10.2025	24.10.2025
4	27.10.2025	28.10.2025	29.10.2025	30.10.2025	31.10.2025
5	03.11.2025	04.11.2025	05.11.2025	06.11.2025	07.11.2025
6	10.11.2025	11.11.2025	12.11.2025	13.11.2025	14.11.2025
7	17.11.2025	18.11.2025	19.11.2025	20.11.2025	21.11.2025
8	24.11.2025	25.11.2025	26.11.2025	27.11.2025	28.11.2025
9	01.12.2025	02.12.2025	03.12.2025	04.12.2025	05.12.2025
10	08.12.2025	09.12.2025	10.12.2025	11.12.2025	12.12.2025
11	15.12.2025	16.12.2025	17.12.2025	18.12.2025	19.12.2025
	22.12.2025	23.12.2025	24.12.2025	25.12.2025	26.12.2025
	29.12.2025	30.12.2025	31.12.2025	01.01.2026	02.01.2026
12	05.01.2026	06.01.2026	07.01.2026	08.01.2026	09.01.2026
13	12.01.2026	13.01.2026	14.01.2026	15.01.2026	16.01.2026
14	19.01.2026	20.01.2026	21.01.2026	22.01.2026	23.01.2026
15	26.01.2026	27.01.2026	28.01.2026	29.01.2026	30.01.2026
16	02.02.2026	03.02.2026	04.02.2026	05.02.2026	06.02.2026

HRO-CIGRE

Samostalan rad

SiP 1. KZ

LV1

LV2

LV3

LV4

LV5

SiP 2. KZ

SiP LV G1
9:15 - 11:30

SiP LV G2
16:15 - 18:30

SiP PRED/AV
9:15 - 13:00

SiP LV G3
13:45 - 16:00

SiP LV G4
16:15 - 18:30

Prezentacija je samo podsjetnik o onome što je obuhvaćeno predavanjem i njezin sadržaj nije dovoljan za učenje!

Obavezna literatura – skripta:

- P. Valožić: Signali i procesi, skripta ,TVZ, 2015
Dostupno preko mojTVZ web stranice predmeta kao 8 pdf dokumenata:

Udžbenici:

- H. Hsu: Schaum's Outline of Signals and Systems
<https://uet.vnu.edu.vn/~tantd/ss/TheoryandProblemsofSS.pdf>
- H. Babić: Signali i sustavi
http://sis.zesoi.fer.hr/predavanja/pdf/sis_2001_skripta.pdf
- B. Jeren: Signali i sustavi
<https://shop.skolskaknjiga.hr/signali-i-sustavi/>
- I.S. Pandžić et al: Uvod u teoriju informacije i kodiranje (3. poglavlje)
<https://element.hr/artikli/293/uvod-u-teoriju-informacije-i-kodiranje>

Navedena literatura je ujedno i izvor za pripremu ove prezentacije.



- Laboratorijske vježbe – prvi dojmovi?
- Ideje zapišite za ZR

Fourierov red u kompleksnom obliku

Progress is the road from the primitive via the complicated to the simple

(Biedenkopf, 1994.)

Polazeći od Fourierovog reda u trigonometrijskom obliku :

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} [a_i \cos(i \cdot \omega_1 t) + b_i \sin(i \cdot \omega_1 t)] \quad i = 1, 2, 3, \dots$$

Uporabom Eulerovih formula za kosinus i sinus:

$$\cos(z) = \frac{1}{2}(e^{jz} + e^{-jz}) \quad \sin(z) = \frac{1}{2j}(e^{jz} - e^{-jz}) \quad z = i \cdot \omega_1 t$$

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} \left[\frac{a_i}{2} (e^{i \cdot j \omega_1 t} + e^{-i \cdot j \omega_1 t}) + \frac{b_i}{2j} (e^{i \cdot j \omega_1 t} - e^{-i \cdot j \omega_1 t}) \right]$$

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} \left[\frac{a_i}{2} (e^{i \cdot j \omega_1 t} + e^{-i \cdot j \omega_1 t}) - \frac{j b_i}{2} (e^{i \cdot j \omega_1 t} - e^{-i \cdot j \omega_1 t}) \right]$$

$$x(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} \left[\left(\frac{a_i - j b_i}{2} \right) e^{i \cdot j \omega_1 t} + \left(\frac{a_i + j b_i}{2} \right) e^{-i \cdot j \omega_1 t} \right]$$

Fourierov red u kompleksnom obliku



$$x(t) = \underbrace{\frac{a_0}{2}}_{c_0} + \sum_{i=1}^{+\infty} \left[\underbrace{\left(\frac{a_i - jb_i}{2} \right)}_{c_i} e^{i \cdot j \omega_1 t} + \underbrace{\left(\frac{a_i + jb_i}{2} \right)}_{c_{-i}} e^{-i \cdot j \omega_1 t} \right] = c_0 + \sum_{i=1}^{+\infty} [c_i \cdot e^{i \cdot j \omega_1 t} + c_{-i} \cdot e^{-i \cdot j \omega_1 t}]$$

Sinteza:

$$x(t) = \sum_{i=-\infty}^{+\infty} c_i \cdot e^{i \cdot j \omega_1 t}$$



Uz a_i i b_i : $a_i = \frac{2}{T} \int_T x(t) \cos(i \cdot \omega_1 t) dt$ $b_i = \frac{2}{T} \int_T x(t) \sin(i \cdot \omega_1 t) dt$

$$a_i, b_i \in \mathbb{R}$$

Dobivamo
za c_i : $c_i = \left(\frac{a_i - jb_i}{2} \right) = \frac{1}{T} \int_T x(t) (\cos(i \cdot \omega_1 t) - j \sin(i \cdot \omega_1 t)) dt$

$$c_i \in \mathbb{C}$$

Primjena Eulerovog identiteta: $\cos(i \cdot \omega_1 t) - j \sin(i \cdot \omega_1 t) = e^{-i \cdot j \omega_1 t}$

Analiza:

$$c_i = \frac{1}{T} \int_T x(t) \cdot e^{-i \cdot j \omega_1 t} dt$$



Iz: $x(t) = \sum_{i=-\infty}^{+\infty} c_i \cdot e^{i \cdot j \omega_1 t}$ $e^{i \cdot j \omega_1 t} = \cos(i \cdot \omega_1 t) + j \sin(i \cdot \omega_1 t)$

$$c_i = \left(\frac{a_i - j b_i}{2} \right), i = 0, 1, 2, \dots \quad c_{-i} = \left(\frac{a_i + j b_i}{2} \right), i = 1, 2, \dots$$



Slijedi:

$$c_{-i} = \overline{c_i} \text{ (konjugirano kompleksna relacija)}$$

$$Re(c_i) = Re(c_{-i}) = a_i/2 \text{ (veza s FR u trigonometrijskom obliku)}$$

$$Im(c_i) = -Im(c_{-i}) = -b_i/2 \text{ (veza s FR u trigonometrijskom obliku)}$$

$$c_i + c_{-i} = a_i$$

$$c_i - c_{-i} = -j b_i$$

PRIMJER: Za realni signal $x(t) = \sum_{i=-\infty}^{+\infty} c_i \cdot e^{i \cdot j \omega_1 t}$ raspis prema **Eulerovoj formuli**:

$$x(t) = c_0 + c_1[\cos(\omega_1 t) + j \sin(\omega_1 t)] + c_{-1}[\cos(-\omega_1 t) + j \sin(-\omega_1 t)] + \\ c_2[\cos(2 \cdot \omega_1 t) + j \sin(2 \cdot \omega_1 t)] + c_{-2}[\cos(-2 \cdot \omega_1 t) + j \sin(-2 \cdot \omega_1 t)] + \dots$$

$$x(t) = c_0 + (c_1 + c_{-1}) \cos(\omega_1 t) + j(c_1 - c_{-1}) \sin(\omega_1 t) + \\ (c_2 + c_{-2}) \cos(2 \cdot \omega_1 t) + j(c_2 - c_{-2}) \sin(2 \cdot \omega_1 t) + \dots$$

$$c_i + c_{-i} = 2\operatorname{Re}(c_i) = a_i$$

$$c_i - c_{-i} = j \cdot 2\operatorname{Im}(c_i) = -jb_i$$

$$a_i, b_i \in \mathbf{R}$$

$$x(t) = c_0 + 2\operatorname{Re}(c_1)\cos(1 \cdot \omega_1 t) - 2\operatorname{Im}(c_1)\sin(1 \cdot \omega_1 t) + \\ 2\operatorname{Re}(c_2)\cos(2 \cdot \omega_1 t) - 2\operatorname{Im}(c_2)\sin(2 \cdot \omega_1 t) + \dots$$

$$x(t) = a_0/2 + a_1\cos(1 \cdot \omega_1 t) + b_1\sin(1 \cdot \omega_1 t) + \\ a_2\cos(2 \cdot \omega_1 t) + b_2\sin(2 \cdot \omega_1 t) + \dots$$

Vrijednosti $x(t) \in \mathbf{R}$

Veza: Trigonometrijski $\leftrightarrow$ kompleksni oblik



$$a_i = 2\operatorname{Re}(c_i) = 2\operatorname{Re}(c_{-i}), i \geq 0$$

$$b_i = -2\operatorname{Im}(c_i) = 2\operatorname{Im}(c_{-i}), i > 0$$

$$c_0 = c_0$$

$$c_i = |2c_i|, i > 0$$

$$\varphi_i = \angle(c_i), i > 0$$



Kalkulatorom, iz kompleksnih koeficijenata c_i u trigonometrijske koeficijente c_i , za $i > 0$:

$$\mathbf{Pol}(2\operatorname{Re}(c_i), 2\operatorname{Im}(c_i)) \rightarrow c_i \angle \varphi_i$$

$$c_i = \left(\frac{a_i - jb_i}{2} \right), i \geq 0$$

$$c_{-i} = \left(\frac{a_i + jb_i}{2} \right), i > 0$$

Kalkulatorom, iz trigonometrijskih koeficijenata c_i u kompleksne koeficijente c_i , za $i > 0$:

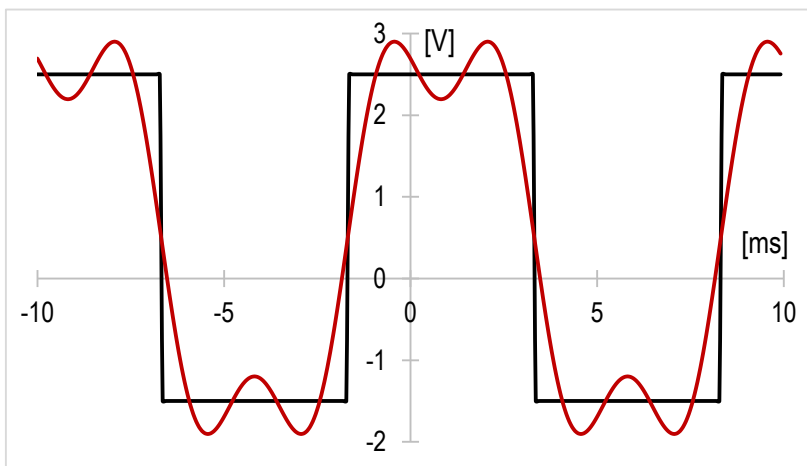
$$\mathbf{Rec}\left(\frac{c_i}{2}, \varphi_i\right) \rightarrow \operatorname{Re}(c_i), \operatorname{Im}(c_i)$$

$$\mathbf{Rec}\left(\frac{c_i}{2}, -\varphi_i\right) \rightarrow \operatorname{Re}(c_{-i}), \operatorname{Im}(c_{-i})$$

Veza: Trigonometrijski $\leftrightarrow$ kompleksni oblik



Primjer: Zadani su koeficijenti Fourierovog reda u kompleksnom obliku c_i za prva tri harmonika pravokutnog signala $f_1 = 100$ Hz. Odredite koeficijente Fourierovog reda u trigonometrijskom obliku.



$$c_{-3} = -0,01043 - 0,42467j$$

$$c_{-1} = 1,10794 + 0,62764j$$

$$c_0 = 0,5$$

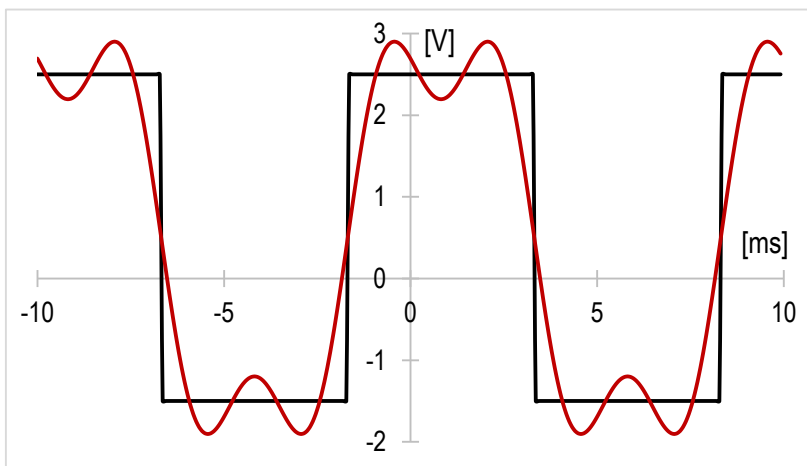
$$c_1 = 1,10794 - 0,62764j$$

$$c_3 = -0,01043 + 0,42467j$$

Veza: Trigonometrijski $\leftrightarrow$ kompleksni oblik



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$$c_3 = -0,01043 + 0,42467j$$

Izračunati koeficijenti Fourierovog reda u trigonometrijskom obliku a_i , b_i i c_i su:

$$a_0 = 1,0$$

$$b_0 = 0,0$$

$$c_0 = 0,5 \angle 0$$

$$a_1 = 2,21588$$

$$b_1 = 1,25528$$

$$c_1 = 2,54673 \angle -0,51542$$

$$a_3 = -0,02085$$

$$b_3 = -0,84934$$

$$c_3 = 0,84959 \angle 1,59534$$

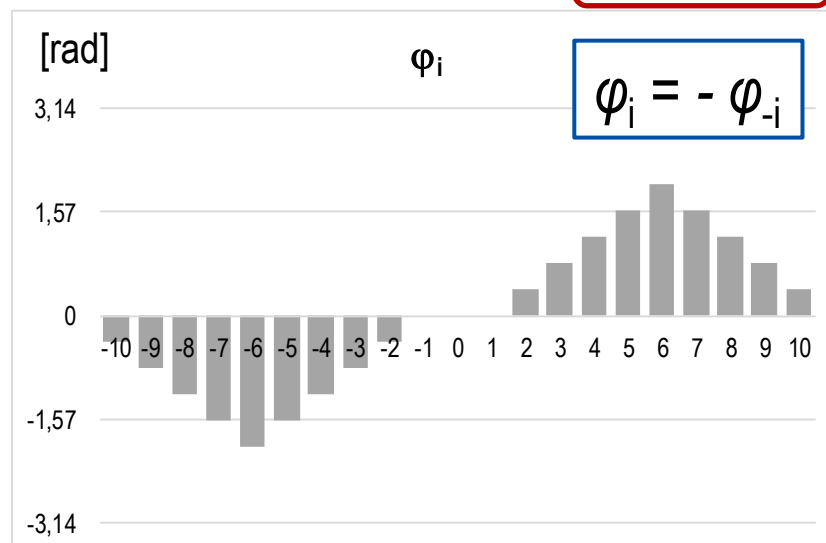
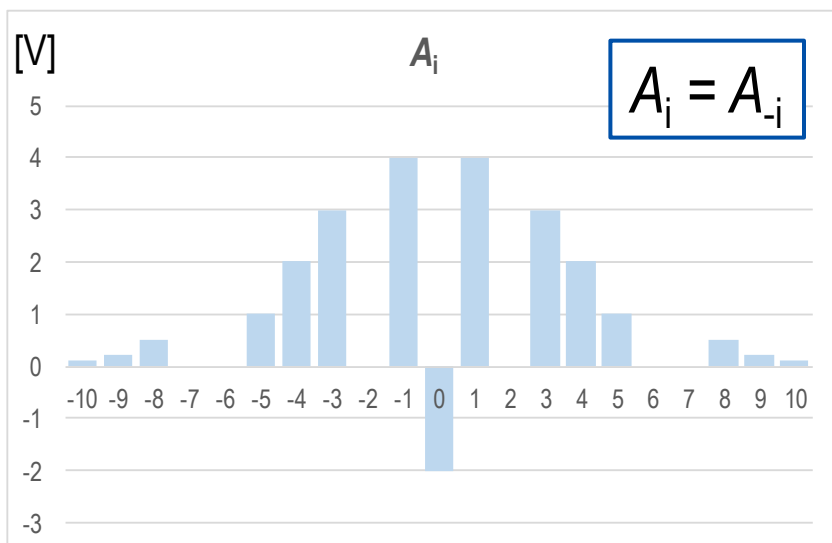
Analitički zapis signala:

$$u(t) = 0,5 + 2,5467 \cdot \cos(2\pi \cdot 100 \cdot t - 0,5154) + 0,8496 \cdot \cos(2\pi \cdot 300 \cdot t + 1,5953) [V]$$

Spektar (realnog) signala



Svaki harmonik složenoga signala prikazan je **konjugirano – kompleksnim** parom polufazora. Jedino tako zbroj kompleksnih brojeva daje **realnu vrijednost** (za realne signale).



$$A_i = |c_i| = \sqrt{\operatorname{Re}(c_i)^2 + \operatorname{Im}(c_i)^2} = \frac{1}{2} \sqrt{a_i^2 + b_i^2}$$

$$\varphi_i = \angle(c_i) = \operatorname{atan2}(\operatorname{Re}(c_i), \operatorname{Im}(c_i))$$

Kalkulatorom, iz kompleksnih koeficijenata c_i u komponente amplitudnog i faznog spektra:

$$\mathbf{Pol}(\operatorname{Re}(c_i), \operatorname{Im}(c_i)) \rightarrow A_i \angle \varphi_i$$



Primjer: Zadani su koeficijenti Fourierovog reda u kompleksnom obliku c_i . Skicirajte amplitudni i fazni spektar signala.

$$c_{-3} = -0,0104 - 0,4247j$$

$$c_{-1} = 1,1079 + 0,6276j$$

$$c_0 = 0,5$$

$$c_1 = 1,1079 - 0,6276j$$

$$c_3 = -0,0104 + 0,4247j$$

Pretvorba iz kompleksnih koeficijenata c_i u komponente amplitudnog i faznog spektra:

$$\mathbf{Pol}(Re(c_i), Im(c_i)) \rightarrow A_i \angle \varphi_i$$

Primjer: Zadani su koeficijenti Fourierovog reda u kompleksnom obliku c_i . Skicirajte amplitudni i fazni spektar signala.

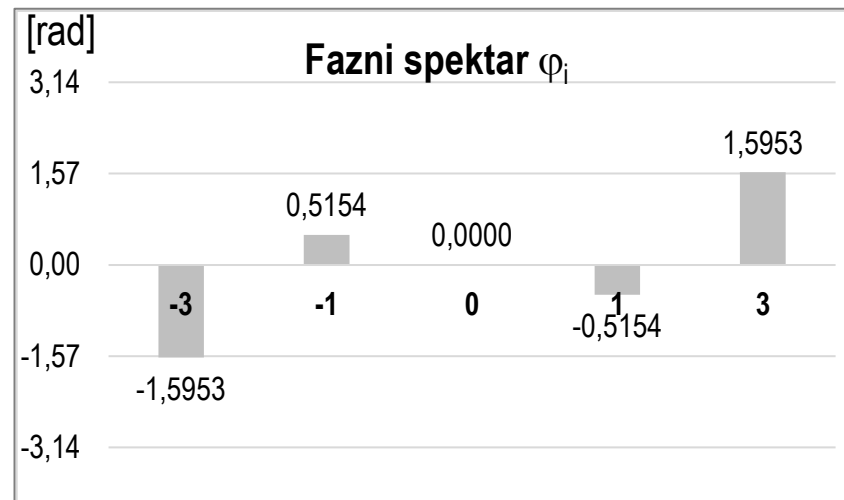
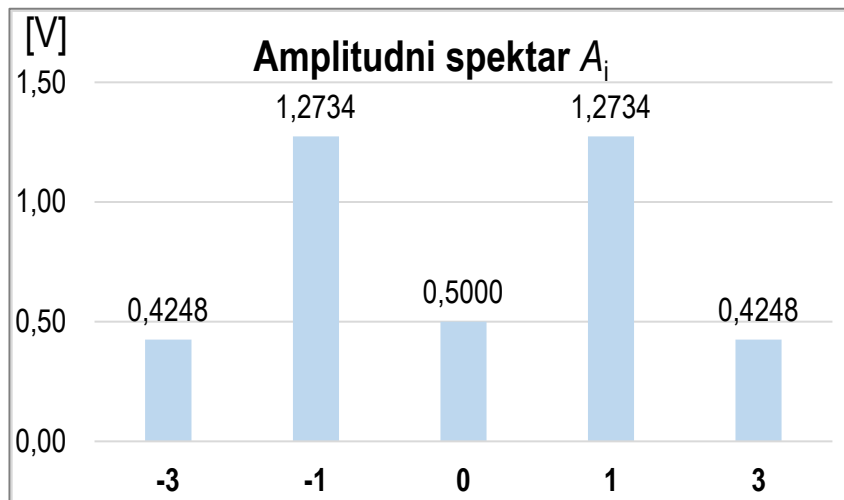
$$c_{-3} = -0,0104 - 0,4247j = \mathbf{0,4248} \angle -1,5953$$

$$c_{-1} = 1,1079 + 0,6276j = \mathbf{1,2734} \angle 0,5154$$

$$c_0 = 0,5 = \mathbf{0,5} \angle 0$$

$$c_1 = 1,1079 - 0,6276j = \mathbf{1,2734} \angle -0,5154$$

$$c_3 = -0,0104 + 0,4247j = \mathbf{0,4248} \angle 1,5953$$



Veza: Analitički ↔ kompleksni oblik



Izračunavanje kompleksnih koeficijenata c_i iz analitičkog zapisa $u(t)$:

$$u(t) = u_0 + \sum u_i \cdot \cos(i \cdot \omega_1 t + \varphi_i), i = 1, 2, \dots$$

$$c_0 = u_0$$

$$c_i = \frac{u_i}{2} \angle \varphi_i \quad c_i = \frac{u_i}{2} \cos \varphi_i + j \frac{u_i}{2} \sin \varphi_i$$

$$c_{-i} = \frac{u_i}{2} \angle -\varphi_i \quad c_{-i} = \frac{u_i}{2} \cos \varphi_i - j \frac{u_i}{2} \sin \varphi_i$$

$$\begin{aligned} & \text{Kalkulatorom, za } i > 0 \\ & \text{Rec}\left(\frac{u_i}{2}, \varphi_i\right) \rightarrow \begin{cases} x = \text{Re}(c_i) \\ y = \text{Im}(c_i) \end{cases} \rightarrow \begin{cases} c_i = \frac{a_i}{2} - j \frac{b_i}{2} \\ c_{-i} = \frac{a_i}{2} + j \frac{b_i}{2} \end{cases} \end{aligned}$$

U slučaju analitičkog zapisa preko sinusa: $u(t) = u_0 + u_i \cdot \sin(i \cdot \omega_1 t + \varphi_i), i = 1, 2, \dots$

Svesti na prethodni oblik (u kosinus): $u(t) = u_0 + u_i \cdot \cos(i \cdot \omega_1 t + \varphi_i - \pi/2)$

Izračunavanje analitičkog zapisa $u(t)$ iz kompleksnih koeficijenata c_i :

Za $i = 0$: $u_0 = c_0$

Za $i > 0$:

$$u_i = |2c_i| = \sqrt{\text{Re}(2c_i)^2 + \text{Im}(2c_i)^2}$$

$$\varphi_i = \angle(c_i) = \text{atan2}(\text{Re}(c_i), \text{Im}(c_i))$$

$$\left. \begin{aligned} & \text{Kalkulatorom:} \\ & \text{Pol}(\text{Re}(2c_i), \text{Im}(2c_i)) \rightarrow u_i \angle \varphi_i \end{aligned} \right\}$$

Veza: Analitički $\leftrightarrow$ kompleksni oblik



Primjer: Zadan je analitički zapis signala $u(t)$. Odredite koeficijente Fourierovog reda u kompleksnom obliku c_i .

$$u(t) = 0,5 + 2,5467 \cdot \cos(2\pi \cdot 100 \cdot t - 0,5154) + 0,8496 \cdot \cos(2\pi \cdot 300 \cdot t + 1,5953) [V]$$

Veza: Analitički ↔ kompleksni oblik



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Kalkulatorom
za $i > 0$: $\text{Rec}\left(\frac{u_i}{2}, \varphi_i\right) \rightarrow \begin{matrix} x = \text{Re}(c_i) \\ y = \text{Im}(c_i) \end{matrix}$

$$c_{-3} = \frac{u_3}{2} \angle -\varphi_3 = 0,4248 \angle -1,5953 = -0,0104 - 0,4247j$$

$$c_{-1} = \frac{u_1}{2} \angle -\varphi_1 = 1,2734 \angle 0,5154 = 1,1079 + 0,6276j$$

$$c_0 = u_0 = 0,5$$

$$c_1 = \frac{u_1}{2} \angle \varphi_1 = 1,2734 \angle -0,5154 = 1,1079 - 0,6276j$$

$$c_3 = \frac{u_3}{2} \angle \varphi_3 = 0,4248 \angle 1,5953 = -0,0104 + 0,4247j$$

Veza: Analitički ↔ kompleksni oblik



Primjer: Zadani su koeficijenti Fourierovog reda u kompleksnom obliku c_i . Odredite analitički zapis signala $u(t)$ ako je $f_1 = 100$ Hz.

$$c_{-3} = -0,0104 - 0,4247j$$

$$c_{-1} = 1,1079 + 0,6276j$$

$$c_0 = 0,5$$

$$c_1 = 1,1079 - 0,6276j$$

$$c_3 = -0,0104 + 0,4247j$$

Pretvorba iz kompleksnih koeficijenata c_i u komponente analitičkog zapisa:

$$\text{za } i = 0: u_0 = c_0$$

$$\text{za } i > 0: \mathbf{Pol}(Re(2c_i), Im(2c_i)) \rightarrow u_i \angle \varphi_i$$

Veza: Analitički ↔ kompleksni oblik



Primjer: Zadani su koeficijenti Fourierovog reda u kompleksnom obliku c_i . Odredite analitički zapis signala $u(t)$ ako je $f_1 = 100$ Hz.

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Pretvorba iz kompleksnih koeficijenata c_i u komponente analitičkog zapisa:

za $i = 0$: $u_0 = c_0$

za $i > 0$: $\text{Pol}(Re(2c_i), Im(2c_i)) \rightarrow u_i \angle \varphi_i$

$$u_0 = 0,5 \text{ V}$$

$$u_1 = 2,5467 \text{ V} \quad \varphi_1 = -0,5154$$

$$u_3 = 0,8496 \text{ V} \quad \varphi_3 = 1,5953$$

$$u(t) = 0,5 + 2,5467 \cdot \cos(2\pi \cdot 100 \cdot t - 0,5154) + 0,8496 \cdot \cos(2\pi \cdot 300 \cdot t + 1,5953) [V]$$

Fourierov red u kompleksnom obliku

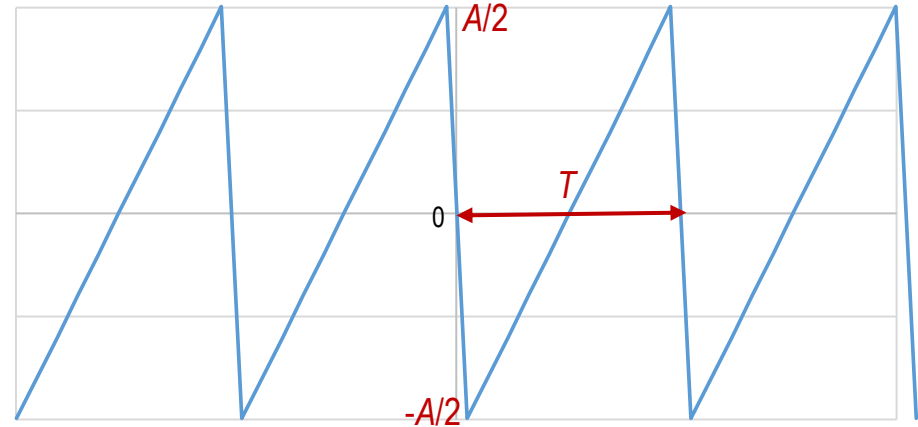
Primjeri signala

Fourierov red – Primjer pilastog signala



$$x(t) = A\left(\frac{t}{T} - \frac{1}{2}\right),$$
$$0 \leq t < T$$

Pilasti signal



$$c_i = \frac{1}{T} \int_0^T x(t) \cdot e^{-i \cdot j \omega_1 t} dt = \frac{1}{T} \int_0^T \left(\frac{At}{T} - \frac{A}{2}\right) \cdot e^{-i \cdot j \omega_1 t} dt$$

$$c_0 = 0$$

$$c_i = \frac{A}{T^2} \int_0^T t \cdot e^{-i \cdot j \omega_1 t} dt - \frac{A}{2T} \int_0^T e^{-i \cdot j \omega_1 t} dt$$

$$c_i = \frac{A}{T^2} \left(\frac{tT}{-i \cdot j \omega_1} e^{-i \cdot j \omega_1 t} + \frac{T^2}{\omega_1^2} e^{-i \cdot j \omega_1 t} \right)_0^T - \frac{A}{2T} \left(\frac{1}{-i \cdot j \omega_1} e^{-i \cdot j \omega_1 t} \right)_0^T$$

Fourierov red – Primjer pilastog signala



$$c_i = \frac{A}{T^2} \left(\frac{tT}{-i \cdot j\omega_1} e^{-i \cdot j\omega_1 t} + \frac{T^2}{\omega_1^2} e^{-i \cdot j\omega_1 t} \right)_0^T - \frac{A}{2T} \left(\frac{1}{-i \cdot j\omega_1} e^{-i \cdot j\omega_1 t} \right)_0^T$$

$$e^{-i \cdot j\omega_1 \cdot 0} = 1 \quad e^{-i \cdot j\omega_1 \cdot T} = 1$$

$$c_i = \frac{A}{T^2} \left(\frac{T^2}{-i \cdot j\omega_1} \right) - \frac{A}{2T} \left(\frac{1}{-i \cdot j\omega_1} - \frac{1}{-i \cdot j\omega_1} \right) = \frac{jA}{i \cdot 2\pi} \quad i = \pm 1, \pm 2, \pm 3, \dots$$

$$c_i = \frac{jA}{i \cdot 2\pi} \Rightarrow A_i = |c_i| = \frac{A}{i \cdot 2\pi}$$

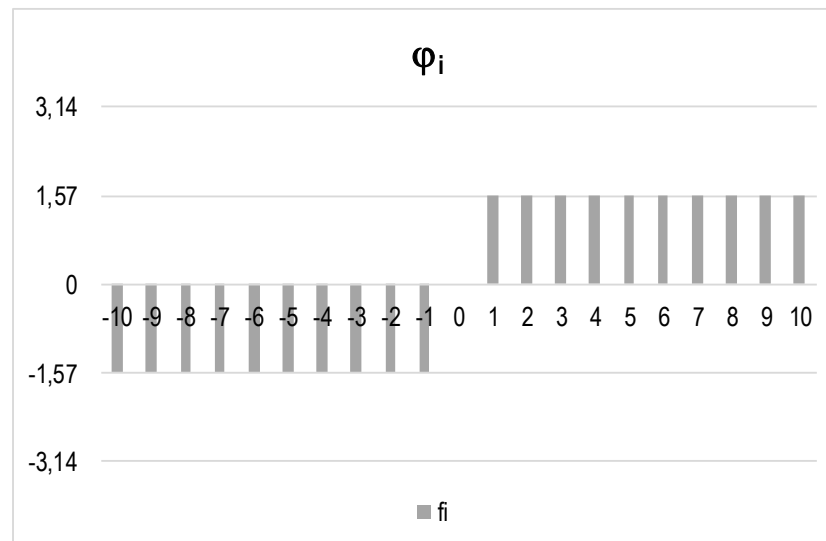
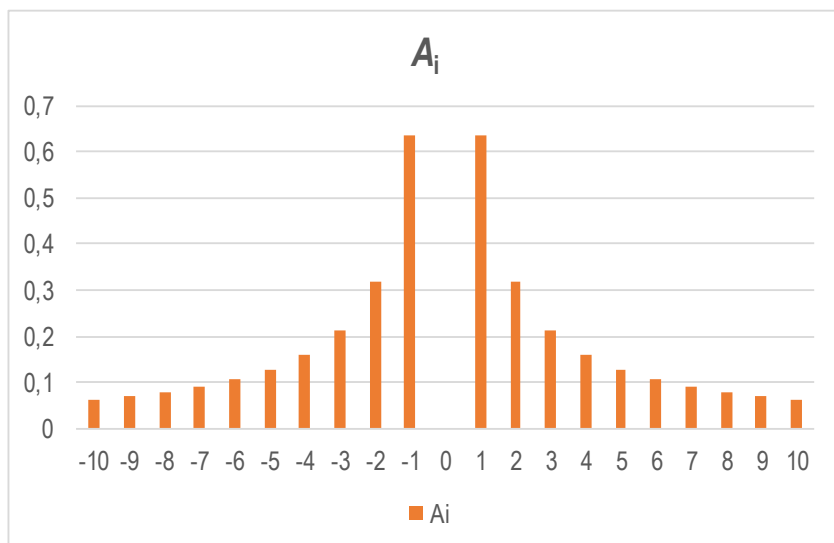
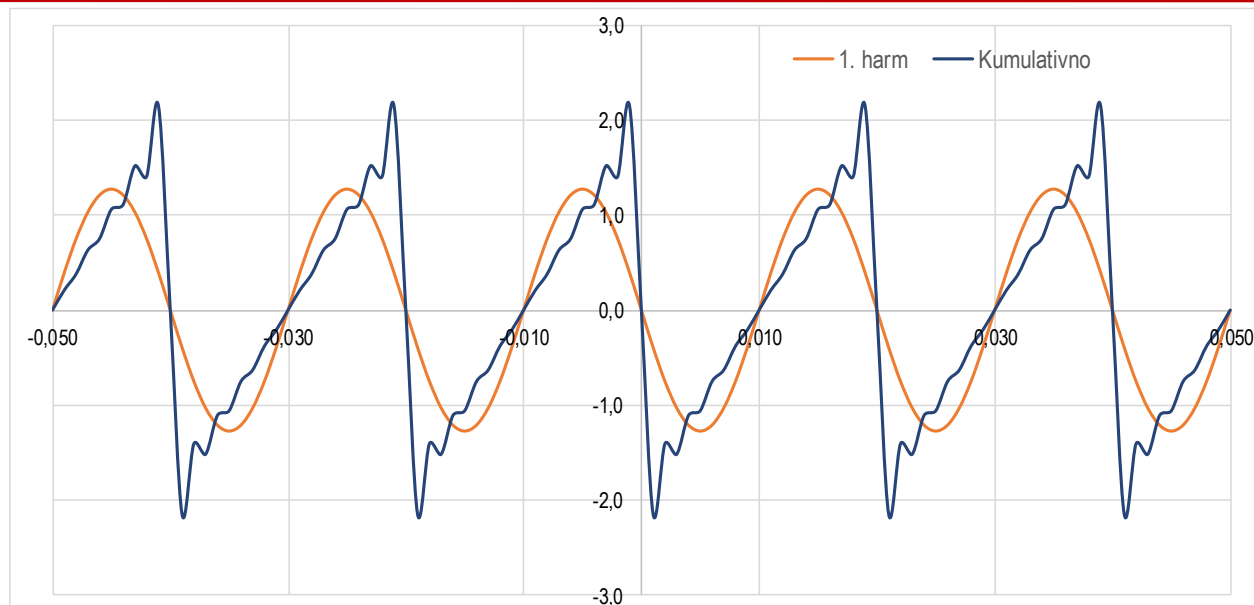
$$\varphi_i = \angle c_i = \pi/2 \quad \text{za } i = +1, +2, +3, \dots$$

$$\varphi_i = \angle c_i = -\pi/2 \quad \text{za } i = -1, -2, -3, \dots$$

Fourierov red – Primjer pilastog signala



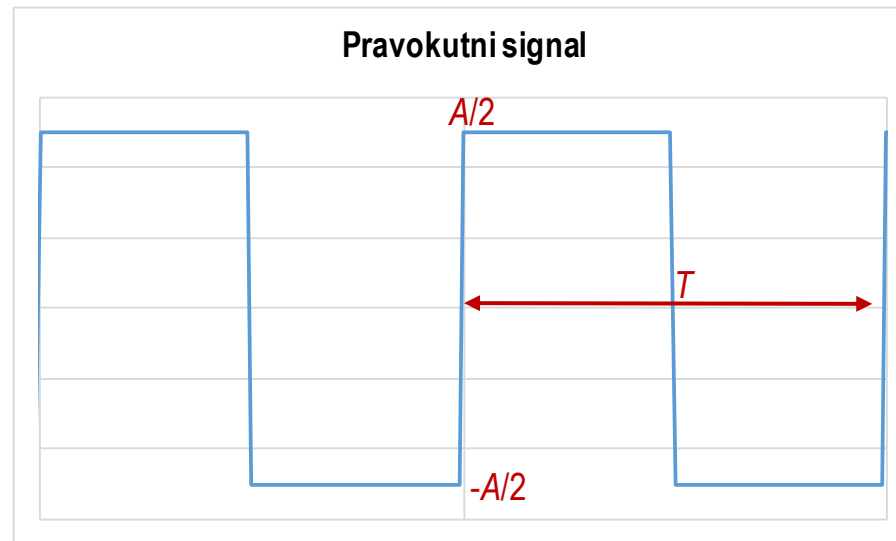
Primjer: $A = 4$, $T = 20$ ms



Fourierov red – Primjer pravokutnog signala



$$x(t) = \begin{cases} +\frac{A}{2} & \text{za } 0 \leq t < T/2 \\ -\frac{A}{2} & \text{za } T/2 \leq t < T \end{cases}$$



$$c_i = -\frac{jA}{i \cdot \pi} \Rightarrow A_i = |c_i| = \frac{A}{i \cdot \pi}$$

$$\varphi_i = \angle c_i = -\pi/2 \quad \text{za } i = +1, +3, +5, \dots$$

$$\varphi_i = \angle c_i = +\pi/2 \quad \text{za } i = -1, -3, -5, \dots$$

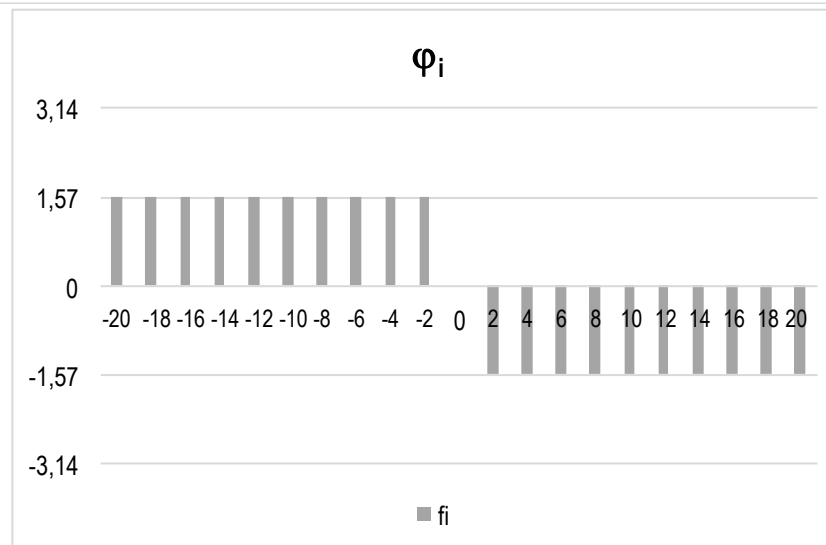
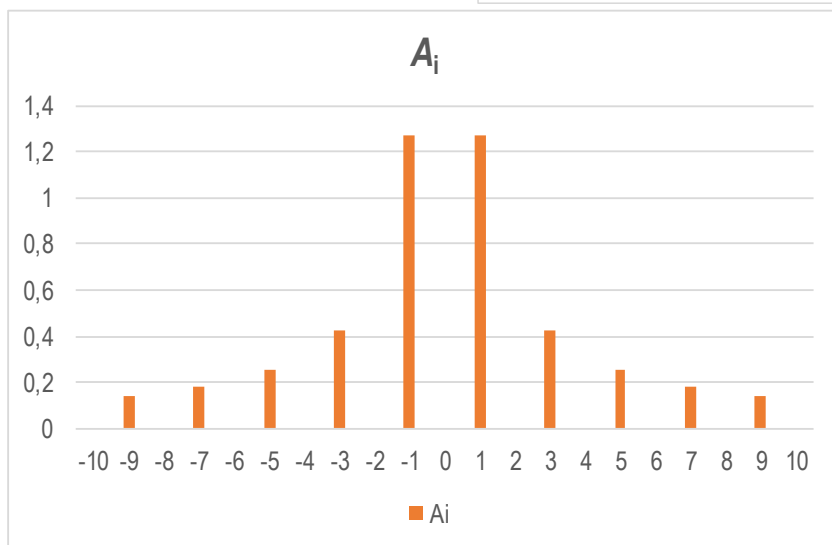
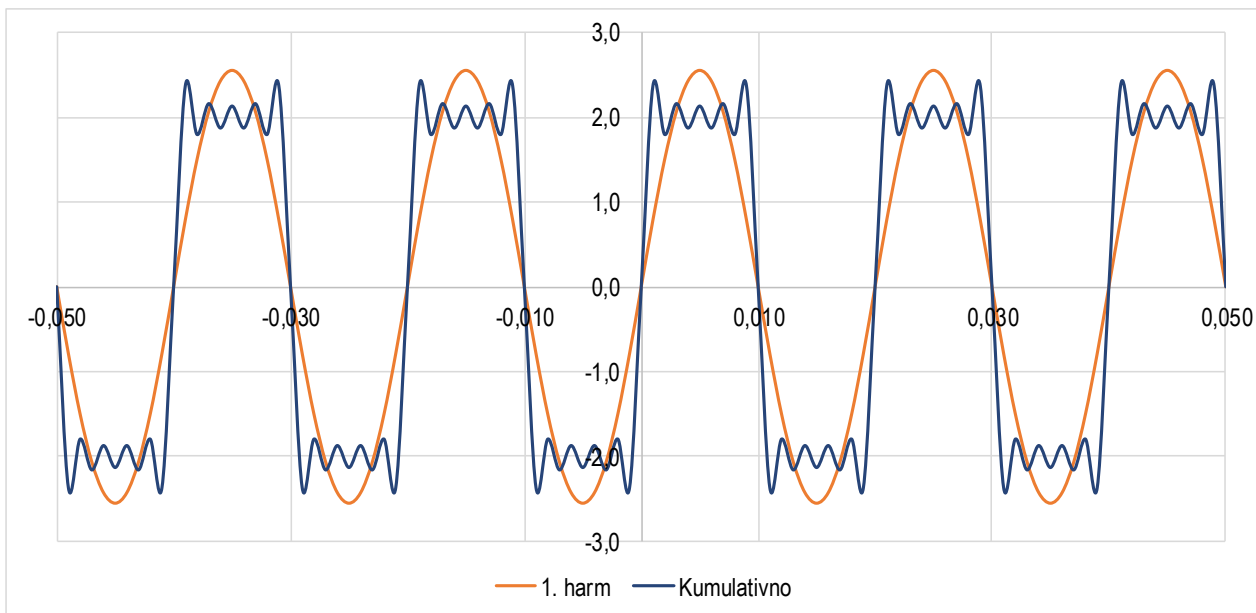
Zzv13-01

Izvedite formule za c_i za
pravokutni signal

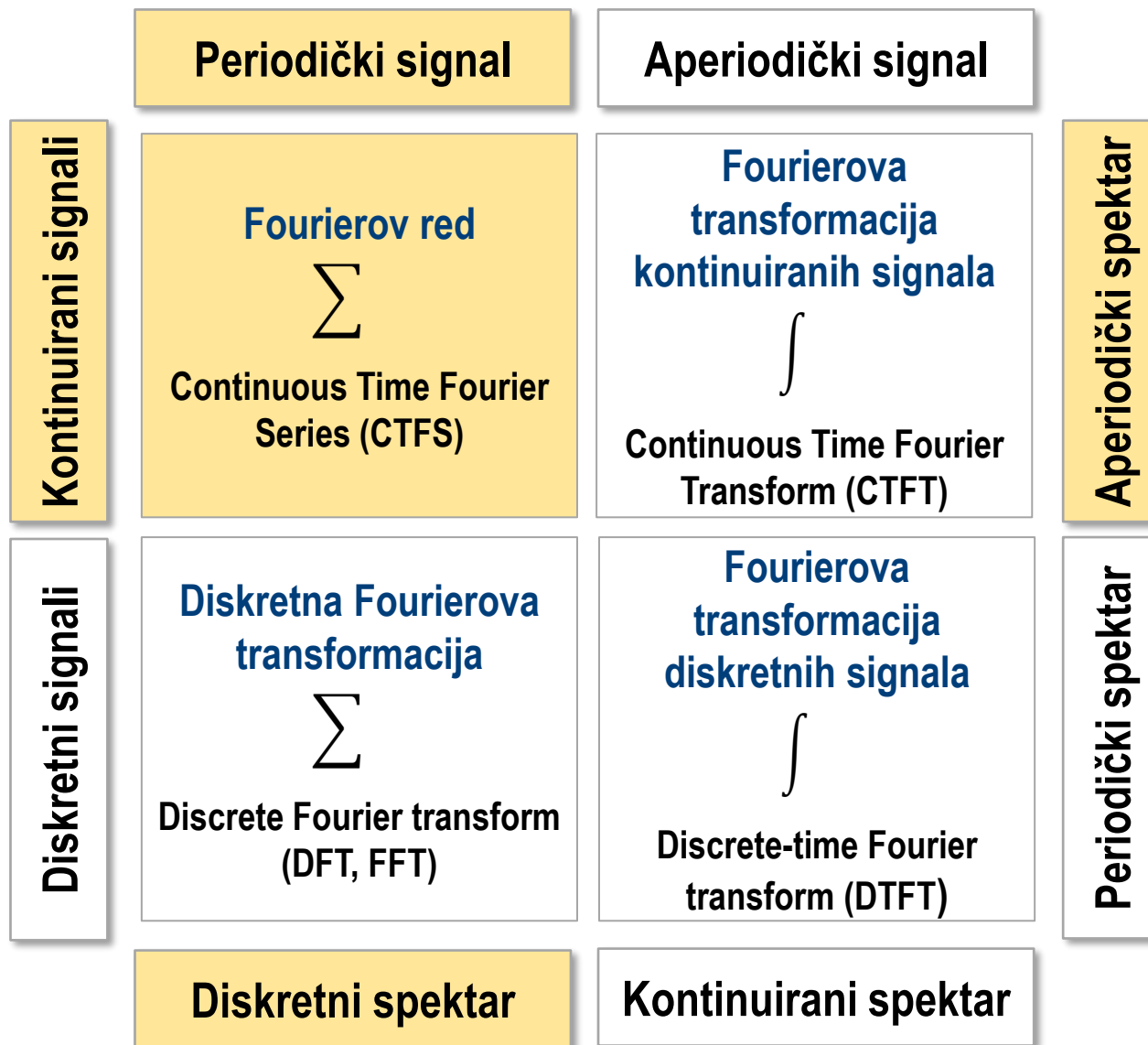
Fourierov red – Primjer pravokutnog signala



Primjer: $A = 4$, $T = 20$ ms



Fourierov red u kompleksnom obliku



Fourierov red u kompleksnom obliku

Zadaci

1. zadatak

Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \sin(\omega_1 t)$

Prema:
$$x(t) = \sum_{i=-\infty}^{+\infty} c_i \cdot e^{i \cdot j \omega_1 t}$$

Tražimo c_1 i c_{-1} , tako da vrijedi: $\sin(\omega_1 t) = c_1 \cdot e^{1 \cdot j \omega_1 t} + c_{-1} \cdot e^{-1 \cdot j \omega_1 t}$

Prema Eulerovom identitetu je: $\sin(\omega_1 t) = -\frac{j}{2} (e^{j \omega_1 t} - e^{-j \omega_1 t})$

Pa slijedi da su: $c_1 = -0,5 \cdot j$ $c_{-1} = +0,5 \cdot j$

Provjera korištenjem Eulerovog identiteta: $e^{i \cdot j \omega_1 t} = \cos(i \cdot \omega_1 t) + j \sin(i \cdot \omega_1 t)$

Općenito za $\underline{x}(t)$: $x(t) = c_1 [\cos(\omega_1 t) + j \sin(\omega_1 t)] + c_{-1} [\cos(-\omega_1 t) + j \sin(-\omega_1 t)]$

$$x(t) = (c_1 + c_{-1}) \cos(\omega_1 t) + j (c_1 - c_{-1}) \sin(\omega_1 t)$$

U ovom slučaju: $u(t) = (0) \cdot \cos(\omega_1 t) + j \cdot (-j) \cdot \sin(\omega_1 t) = \sin(\omega_1 t)$

1. zadatak – jednostavnije!

Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \sin(\omega_1 t)$

Prema:

$$c_i = \left(\frac{a_i - jb_i}{2} \right), i = 1, 2, \dots$$

$$c_{-i} = \left(\frac{a_i + jb_i}{2} \right), i = 1, 2, \dots$$

$$u(t) = a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t)$$

$$a_1 = 0$$

$$b_1 = 1$$

Sijedi da su: $c_1 = -0,5 \cdot j$ $c_{-1} = +0,5 \cdot j$

Provjerite uvrštavanjem:

$$x(t) = \sum_{i=-1}^{+1} c_i \cdot e^{i \cdot j \omega_1 t} = c_1 [\cos(i \cdot \omega_1 t) + j \sin(i \cdot \omega_1 t)] + c_{-1} [\cos(i \cdot \omega_1 t) - j \sin(i \cdot \omega_1 t)]$$

2. zadatak – Samostalno!



Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \cos(\omega_1 t)$

2. zadatak

Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \cos(\omega_1 t)$

Prema:
$$x(t) = \sum_{i=-\infty}^{+\infty} c_i \cdot e^{i \cdot j \omega_1 t}$$

Tražimo c_1 i c_{-1} , tako da vrijedi: $\cos(\omega_1 t) = c_1 \cdot e^{1 \cdot j \omega_1 t} + c_{-1} \cdot e^{-1 \cdot j \omega_1 t}$

Prema Eulerovom identitetu je: $\cos(\omega_1 t) = \frac{1}{2} (e^{j \omega_1 t} + e^{-j \omega_1 t})$

Pa slijedi da su: $c_1 = +0,5$ $c_{-1} = +0,5$

Provjera korištenjem Eulerovog identiteta: $e^{i \cdot j \omega_1 t} = \cos(i \cdot \omega_1 t) + j \sin(i \cdot \omega_1 t)$

Općenito za $\underline{x}(t)$:
$$x(t) = c_1 [\cos(\omega_1 t) + j \sin(\omega_1 t)] + c_{-1} [\cos(-\omega_1 t) + j \sin(-\omega_1 t)]$$

$$x(t) = (c_1 + c_{-1}) \cos(\omega_1 t) + j (c_1 - c_{-1}) \sin(\omega_1 t)$$

U ovom slučaju: $u(t) = (1) \cdot \cos(\omega_1 t) + j \cdot (0) \cdot \sin(\omega_1 t) = \cos(\omega_1 t)$

2. zadatak – jednostavnije!



Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \cos(\omega_1 t)$

Prema:

$$c_i = \left(\frac{a_i - jb_i}{2} \right), i = 1, 2, \dots$$

$$c_{-i} = \left(\frac{a_i + jb_i}{2} \right), i = 1, 2, \dots$$

Samostalno!

2. zadatak – jednostavnije!



Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = \cos(\omega_1 t)$

Prema:

$$c_i = \left(\frac{a_i - jb_i}{2} \right), i = 1, 2, \dots$$

$$c_{-i} = \left(\frac{a_i + jb_i}{2} \right), i = 1, 2, \dots$$

$$u(t) = a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t)$$

$$a_1 = 1$$

$$b_1 = 0$$

Sijedi da su: $c_1 = +0,5$ $c_{-1} = +0,5$

Provjerite uvrštavanjem:

$$x(t) = \sum_{i=-1}^{+1} c_i \cdot e^{i \cdot j \omega_1 t} = 2 \left(\frac{1}{2} \right) \cdot \cos(\omega_1 t) + j \cdot (0) \cdot \sin(\omega_1 t) = \cos(\omega_1 t)$$

3. zadatak

Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = 5 + 4 \cdot \cos(100 \cdot t) + 3 \cdot \sin(200 \cdot t)$. Na temelju rezultata, izračunajte i skicirajte komponente amplitudnog i faznog spektra.

Rješenje:

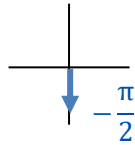
$$c_i = \left(\frac{a_i - jb_i}{2} \right), i = 1, 2, \dots \quad c_{-i} = \left(\frac{a_i + jb_i}{2} \right), i = 1, 2, \dots$$

$$i = 0, \pm 1, \pm 2$$

$$c_0 = 5$$

$$c_1 = 2 \quad c_{-1} = 2$$

$$c_2 = -1,5j \quad c_{-2} = 1,5j$$


 $\sin(x) = \cos\left(x - \frac{\pi}{2}\right) \Rightarrow \angle -\pi/2 = -j \Rightarrow c_2 = -1,5j$

Provjera:

$$x(t) = c_0 + 2\operatorname{Re}(c_1)\cos(1 \cdot \omega_1 t) - 2\operatorname{Im}(c_1)\sin(1 \cdot \omega_1 t) + 2\operatorname{Re}(c_2)\cos(2 \cdot \omega_1 t) - 2\operatorname{Im}(c_2)\sin(2 \cdot \omega_1 t) + \dots$$

$$u(t) = 5 + 4 \cdot \cos(100 \cdot t) + 3 \cdot \sin(200 \cdot t)$$

$$u(t) = 5 + 4 \cdot \cos(100 \cdot t) + 3 \cdot \cos(200 \cdot t - \pi/2)$$

3. zadatak (nastavak)

$$u(t) = 5 + 4 \cdot \cos(100 \cdot t) + 3 \cdot \sin(200 \cdot t)$$

Koeficijenti Fourierovog reda u kompleksnom obliku:

$$c_0 = 5$$

$$c_1 = 2$$

$$c_2 = -1,5j$$

$$c_{-1} = 2$$

$$c_{-2} = 1,5j$$

$$\text{Pol}(\text{Re}(c_i), \text{Im}(c_i)) \Rightarrow A_i, \varphi_i$$

Amplitude spektra Fourierovog reda u kompleksnom obliku:

$$A_0 = c_0 = 5$$

$$A_1 = |c_1| = |2| = 2$$

$$A_2 = |c_2| = |-1,5j| = 1,5$$

$$A_{-1} = |c_{-1}| = |2| = 2$$

$$A_{-2} = |c_{-2}| = |1,5j| = 1,5$$

Faze spektra Fourierovog reda u kompleksnom obliku:

$$\varphi_0 = 0$$

$$\varphi_1 = 0$$

$$\varphi_2 = -\pi/2$$

$$\varphi_{-1} = 0$$

$$\varphi_{-2} = +\pi/2$$

4. zadatak



Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = 4 + 3 \cdot \cos(100 \cdot t + \pi/4) + 2 \cdot \cos(200 \cdot t + \pi/2) + \cos(300 \cdot t + 3\pi/4)$

Na temelju rezultata, izračunajte i skicirajte komponente amplitudnog i faznog spektra.

4. zadatak - rješenje

Za analitički oblik: $u(t) = u_0 + u_i \cdot \cos(i \cdot \omega_1 t + \varphi_i)$, $i = 1, 2, \dots$

Koeficijenti Fourierovog reda u kompleksnom obliku c_i su:

$$u_i \cos(i \cdot \omega t + \varphi_i) \Rightarrow c_i = \frac{u_i}{2} [\cos(\varphi_i) + j \sin(\varphi_i)], i > 0$$

$$c_0 = u_0$$

$$c_{-i} = \overline{c_i}$$

Kalkulatorom: $\text{Rec}\left(\frac{u_i}{2}, \varphi_i\right) \rightarrow \begin{matrix} x = \text{Re}(c_i) \\ y = \text{Im}(c_i) \end{matrix}$

$$u(t) = 4 + 3 \cdot \cos(100 \cdot t + \pi/4) + 2 \cdot \cos(200 \cdot t + \pi/2) + \cos(300 \cdot t + 3\pi/4)$$

$$c_0 = 4$$

$$c_1 = 1,06066 + 1,06066j$$

$$c_2 = j$$

$$c_3 = -0,35355 + 0,35355j$$

$$c_{-1} = 1,06066 - 1,06066j$$

$$c_{-2} = -j$$

$$c_{-3} = -0,35355 - 0,35355j$$

4. zadatak - rješenje



$$u(t) = 4 + 3 \cdot \cos(100 \cdot t + \pi/4) + 2 \cdot \cos(200 \cdot t + \pi/2) + \cos(300 \cdot t + 3\pi/4)$$

Komponente amplitudnog i faznog spektra

Amplitude spektra Fourierovog reda u kompleksnom obliku:

$$A_0 = 4$$

$$A_1 = 1,5 \quad A_{-1} = 1,5$$

$$A_2 = 1 \quad A_{-2} = 1$$

$$A_3 = 0,5 \quad A_{-3} = 0,5$$

Faze spektra Fourierovog reda u kompleksnom obliku:

$$\varphi_0 = 0$$

$$\varphi_1 = \pi/4 \quad \varphi_{-1} = -\pi/4$$

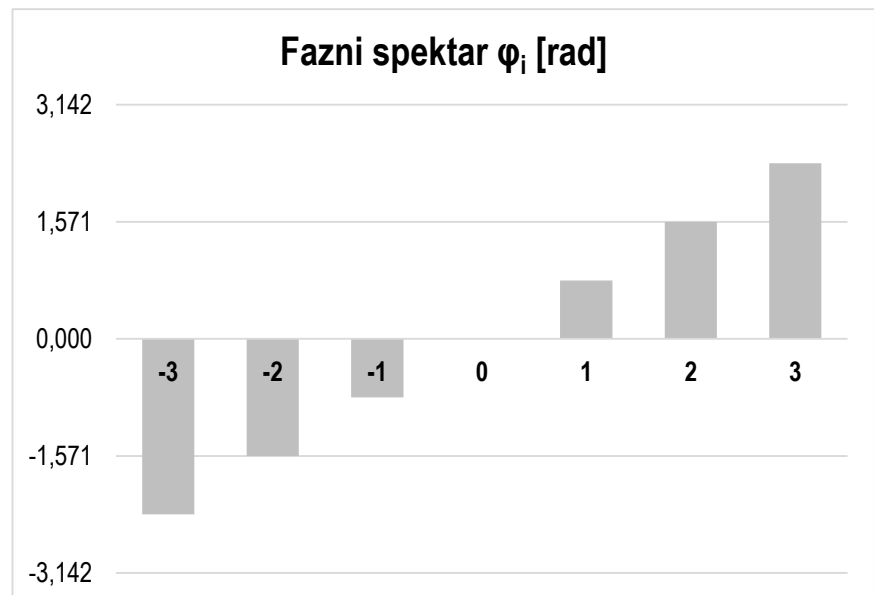
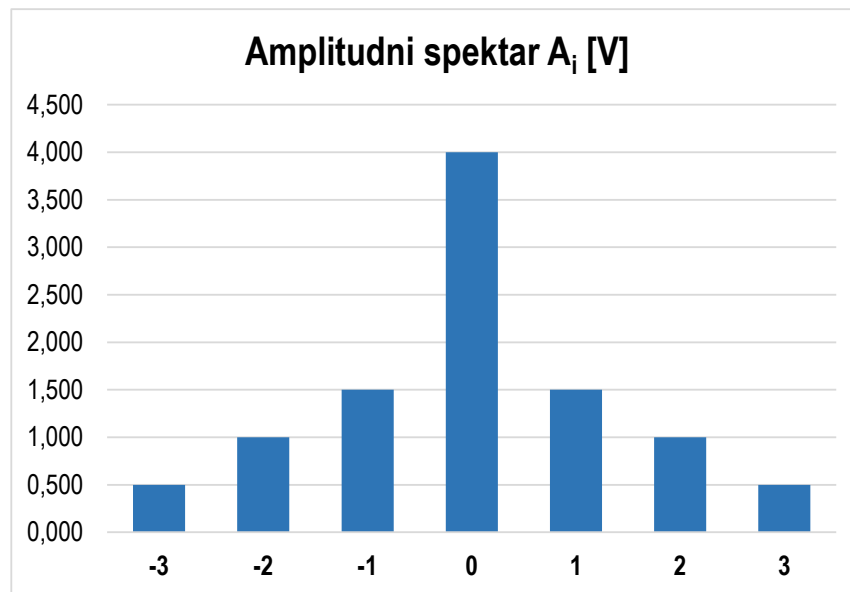
$$\varphi_2 = \pi/2 \quad \varphi_{-2} = -\pi/2$$

$$\varphi_3 = 3\pi/4 \quad \varphi_{-3} = -3\pi/4$$

4. zadatak - rješenje



$$u(t) = 4 + 3 \cdot \cos(100 \cdot t + \pi/4) + 2 \cdot \cos(200 \cdot t + \pi/2) + \cos(300 \cdot t + 3\pi/4)$$



5. Zadatak – Samostalno!



Izračunajte i skicirajte komponente amplitudnog i faznog spektra za signal $u(t)$ kojemu su zadane vrijednosti koeficijenta c_i Fourierovog reda u kompleksnom obliku. Zapišite analitički oblik signala ako je osnovna frekvencija $f_1 = 100$ Hz.

$$c_0 = 5$$

$$c_1 = 0 - j$$

$$c_2 = 2 + j$$

$$c_3 = -1 - 2j$$

5. zadatak - rješenje

Izračunajte i skicirajte komponente amplitudnog i faznog spektra za signal $u(t)$ kojemu su zadane vrijednosti koeficijenta c_i Fourierovog reda u kompleksnom obliku. Zapišite analitički oblik signala ako je osnovna frekvencija $f_1 = 100$ Hz.

$$c_0 = 5$$

$$c_1 = 0 - j$$

$$c_2 = 2 + j$$

$$c_3 = -1 - 2j$$

$$\Rightarrow c_{-1} = 0 + j$$

$$\Rightarrow c_{-2} = 2 - j$$

$$\Rightarrow c_{-3} = -1 + 2j$$

Pol($\text{Re}(c_i)$, $\text{Im}(c_i)$) $\Rightarrow A_i$, φ_i

$$A_0 = 5$$

$$A_1 = A_{-1} = 1$$

$$A_2 = A_{-2} = 2,236$$

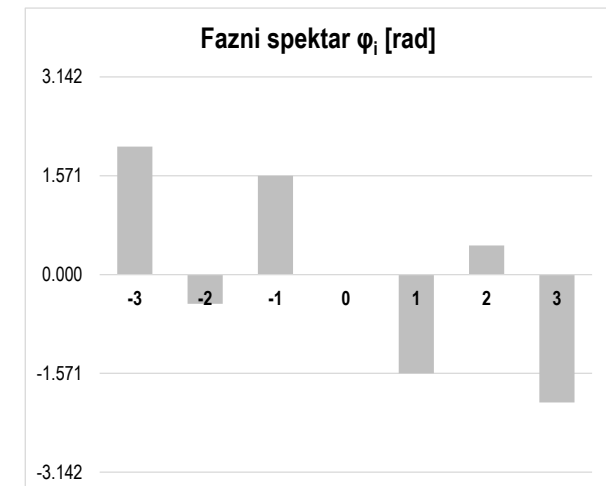
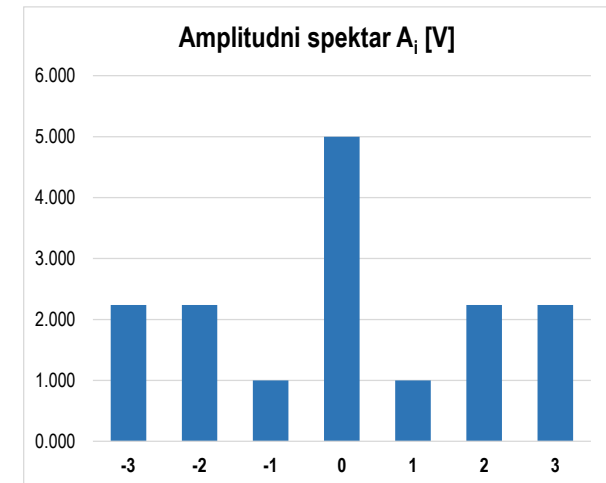
$$A_3 = A_{-3} = 2,236$$

$$\varphi_1 = -\varphi_{-1} = -\pi/2$$

$$\varphi_2 = -\varphi_{-2} = 0,4636$$

$$\varphi_3 = -\varphi_{-3} = -2,0344$$

$$u(t) = 5 + 2\cos(200\pi \cdot t - \pi/2) + \\ 4,4721\cos(400\pi \cdot t + 0,4636) + \\ 4,4721\cos(600\pi \cdot t - 2,0344)$$



6. zadatak

Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = 1 + 2 \cdot \cos(100 \cdot t + 1) + 3 \cdot \sin(100 \cdot t + 2)$. Na temelju rezultata, izračunajte i skicirajte komponente amplitudnog i faznog spektra.

Rješenje:

Potrebno je fazorski zbrojiti $\cos$ i $\sin$ s ciljem dobivanja $\cos$!

$$\text{Rec}(2, 1) = 1,0806 + 1,6829j$$

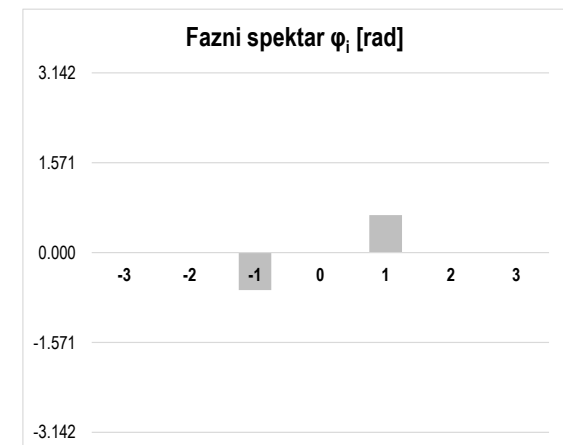
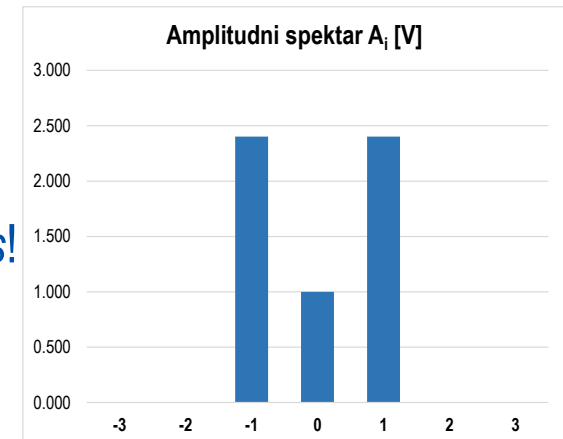
$$\text{Rec}(3, 2 - \pi/2) = \underline{2,7279 + 1,2484j}$$

$$\text{Zbroj} = 3,8086 + 2,9313j$$

$$\text{Pol}(3,8086, 2,9313) = \mathbf{4,8060} \angle \mathbf{0,6560}$$

$$\mathbf{4,8060} \Rightarrow A_1 = 4,8060/2 = 2,4030 = A_{-1}$$

$$\mathbf{0,6560} \Rightarrow \varphi_1 = 0,6560; \varphi_{-1} = -0,6560$$



Odredite koeficijente c_i Fourierovog reda u kompleksnom obliku za signal: $u(t) = 6 + 3 \cdot \cos(100 \cdot t - \pi/3) + 2 \cdot \cos(200 \cdot t - \pi/2) + \cos(300 \cdot t + 2\pi/3)$

Na temelju rezultata, izračunajte i skicirajte komponente amplitudnog i faznog spektra.

Provjerite točnost dobivenih koeficijenata uvrštavanjem u formulu:

$$x(t) = c_0 + 2\operatorname{Re}(c_1)\cos(1 \cdot \omega_1 t) - 2\operatorname{Im}(c_1)\sin(1 \cdot \omega_1 t) + 2\operatorname{Re}(c_2)\cos(2 \cdot \omega_1 t) - 2\operatorname{Im}(c_2)\sin(2 \cdot \omega_1 t) + \dots$$

Parsevalov teorem

za Fourierov red u kompleksnom obliku

Parsevalov teorem za Fourierov red u kompleksnom obliku



Parsevalov teorem

- Udio jedne komponente u ukupnoj snazi složenog signala proporcionalan je kvadratu amplitude te komponente.
- Ukupna snaga signala jednaka je sumi snaga pojedinačnih komponenti ili integralu kvadrata vremenske funkcije signala.

$$\sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{T} \int_T [x(t)]^2 dt$$

Zzv13-03

U usporedbi s Parsevalovim teoremom za FR u trigonometrijskom obliku, gdje je nestala $\frac{1}{2}$ ispred kvadrata c_i ?



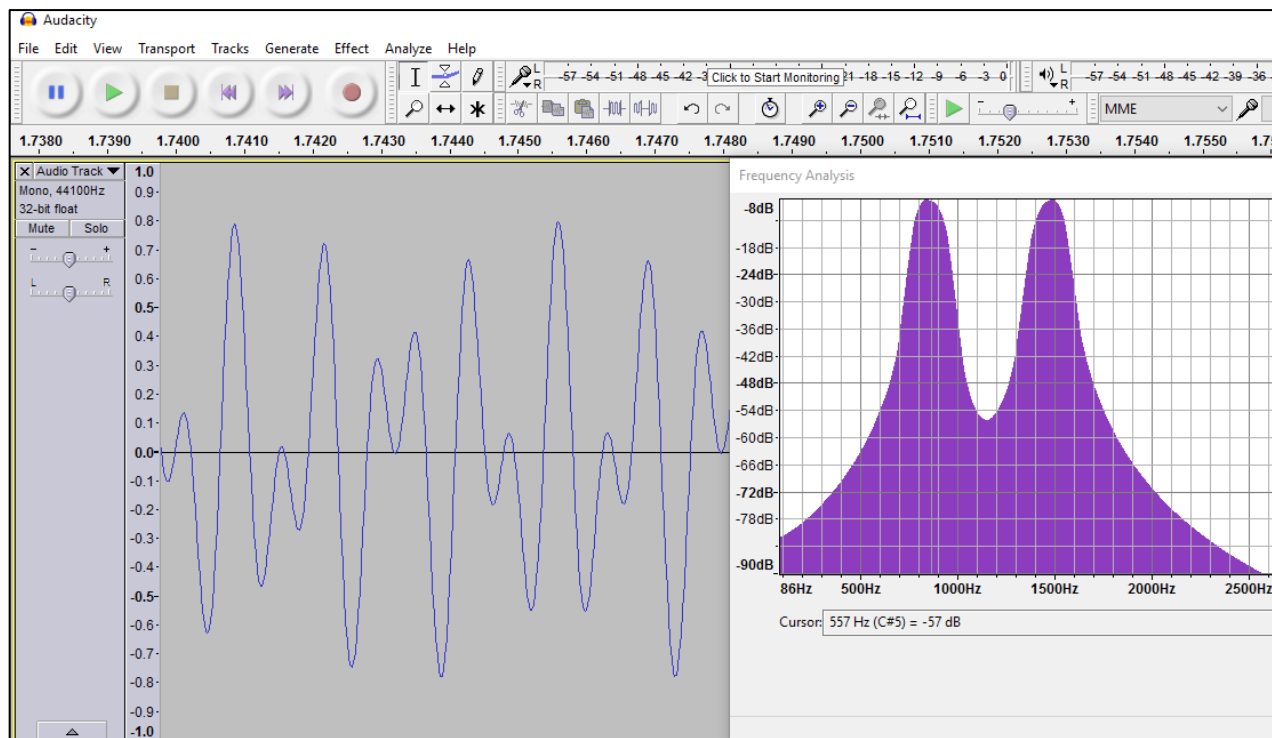
$$P_{sr} = \frac{1}{R} \sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{RT} \int_T [x(t)]^2 dt = \frac{U_{ef}^2}{R}$$

U programu [Audacity](#).

- Generirajte DTMF tonove za brojke 0-9: [Generate / DTMF Tones](#)
- Otvorite prikaz frekvencijskog spektra: [Analyze / Plot Spectrum](#)

Što točnije odredite od kojih frekvencije se sastoji navedeni tonovi.

Za brojku 9 skicirajte (pojedinačne) signale koje čine navedeni ton u vremenskoj domeni.



Ispitajte točnost Parsevalovog teorema na primjeru **Fourierovog reda** u **kompleksnom obliku** za **pravokutni signal** :

$$\boxed{\sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{T} \int_T [x(t)]^2 dt} \quad |c_i| = \frac{A}{i \cdot \pi} \quad i = \pm 1, \pm 3, \pm 5, \dots$$

Desna strana: Koliko iznosi P_{sr} pravokutnog signala navedene amplitude?

Lijeva strana: Za izračunavanje sume $|c_i|^2$ koristite identitet:

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

Ispitajte točnost Parsevalovog teorema na primjeru **Fourierovog reda** u kompleksnom obliku za pilasti signal:

$$\boxed{\sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{T} \int_T [x(t)]^2 dt} \quad |c_i| = \frac{A}{i \cdot 2\pi} \quad i = \pm 1, \pm 2, \pm 3, \dots$$

Desna strana: Koliko iznosi P_{sr} pilastog signala navedene amplitude?

Lijeva strana: Za izračunavanje sume $|c_i|^2$ koristite identitet:

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$